A CONNECTED COUNTEREXAMPLE $K_{3,3}$ TO THE AIM BALANCED-CIRCULANT CONJECTURE AND THE CORRECTED CLASSIFICATION

ARGUS OPEN RESEARCH PROJECT

Abstract. Section K.1 of the AIM perfect-graph problem list conjectures that every nonempty balanced circulant graph is isomorphic to $G_B(k,m) = C_{4m}[\overline{K_k}]$ on $4km$ vertices. We verify that the connected circulant $K_{3,3} = \mathrm{Circ}(6;\{1,3\})$ on 6 vertices is balanced (all 9 induced cycles have length 4) yet cannot have $4km$ vertices, and confirm by exhaustive enumeration of all 196,573 nonempty circulants of order $n \le 32$ that the 106 balanced circulants agree exactly with the Morris–Spiga–Webb classification ($G_B(k,m)$ with $m \ge 2$ or $K_{t,t}$ with $t \ge 1$ on each connected component).

1. Main Proposition

Theorem 1.1. *The circulant graph $G = \mathrm{Circ}(6;\{1,3\})$ on $\mathbb{Z}_6$ satisfies:*

(1) *G is connected, 3-regular, vertex-transitive, and bipartite with parts $\{0,2,4\}$ and $\{1,3,5\}$, hence isomorphic to $K_{3,3}$.*
(2) *Every induced cycle of G has length $4 \equiv 0 \pmod 4$ (with zero induced C_6), so G is balanced.*
(3) *Since $|V(G)| = 6$ is not divisible by 4, G is not isomorphic to $G_B(k,m)$ for any positive integers k, m.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001159 in the domain of algebraic graph theory and perfect graphs follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: K_{3,3} = Circ(6;{1,3}) is connected, balanced, and circulant on 6 != 4km vertices; all 196,573 circulants of order <=32 verified.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 20001159 (algebraic graph theory and perfect graphs), archived with exact theorem locators and verification certificates in the accompanying repository package.

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